Sreelaja Nair

The Immune System

1. Multiple layers of protection
2. Prevent
3. Eliminate

Innate immunity = all animals and plants

Adaptive immunity = only in vertebrates

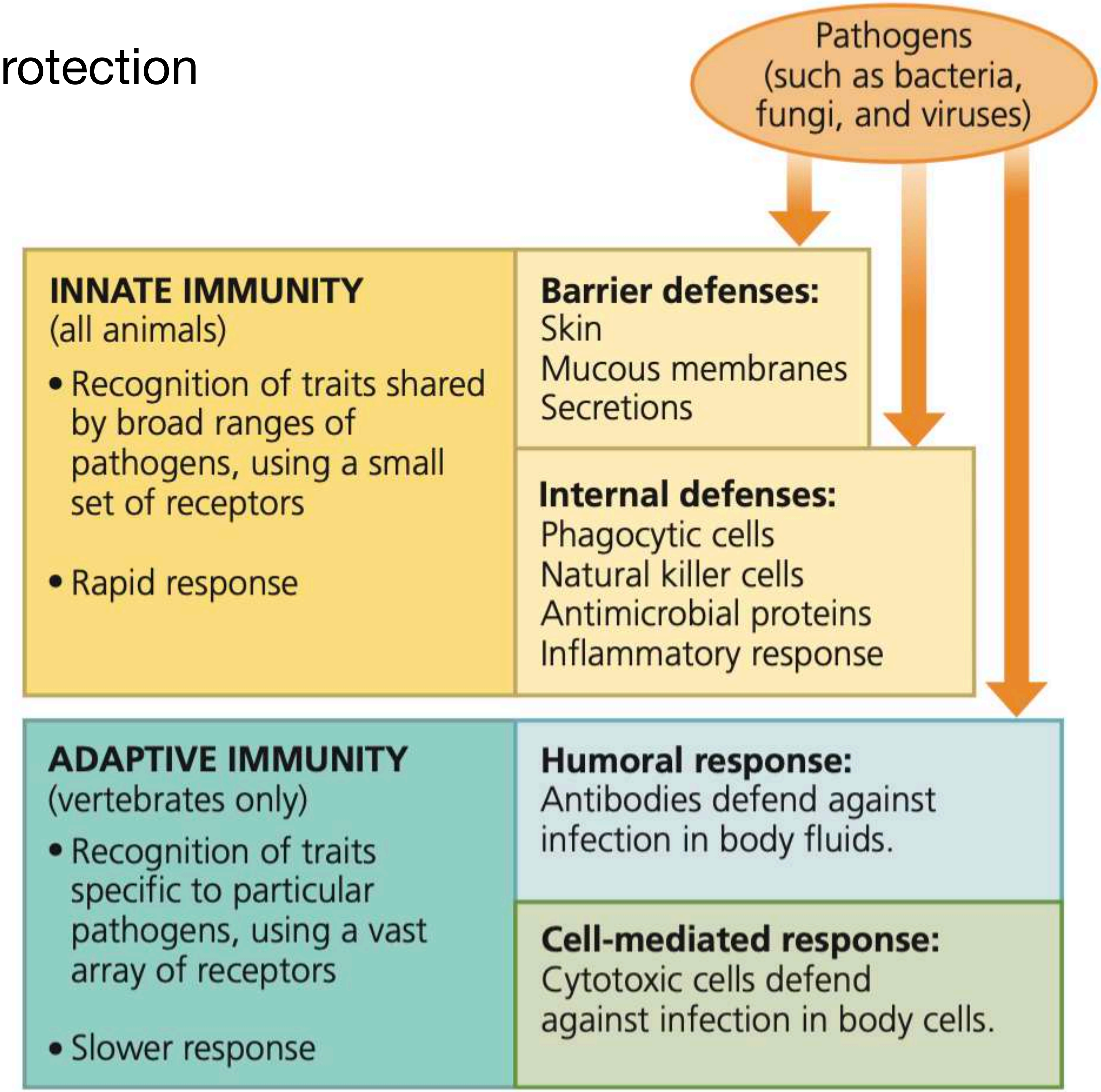
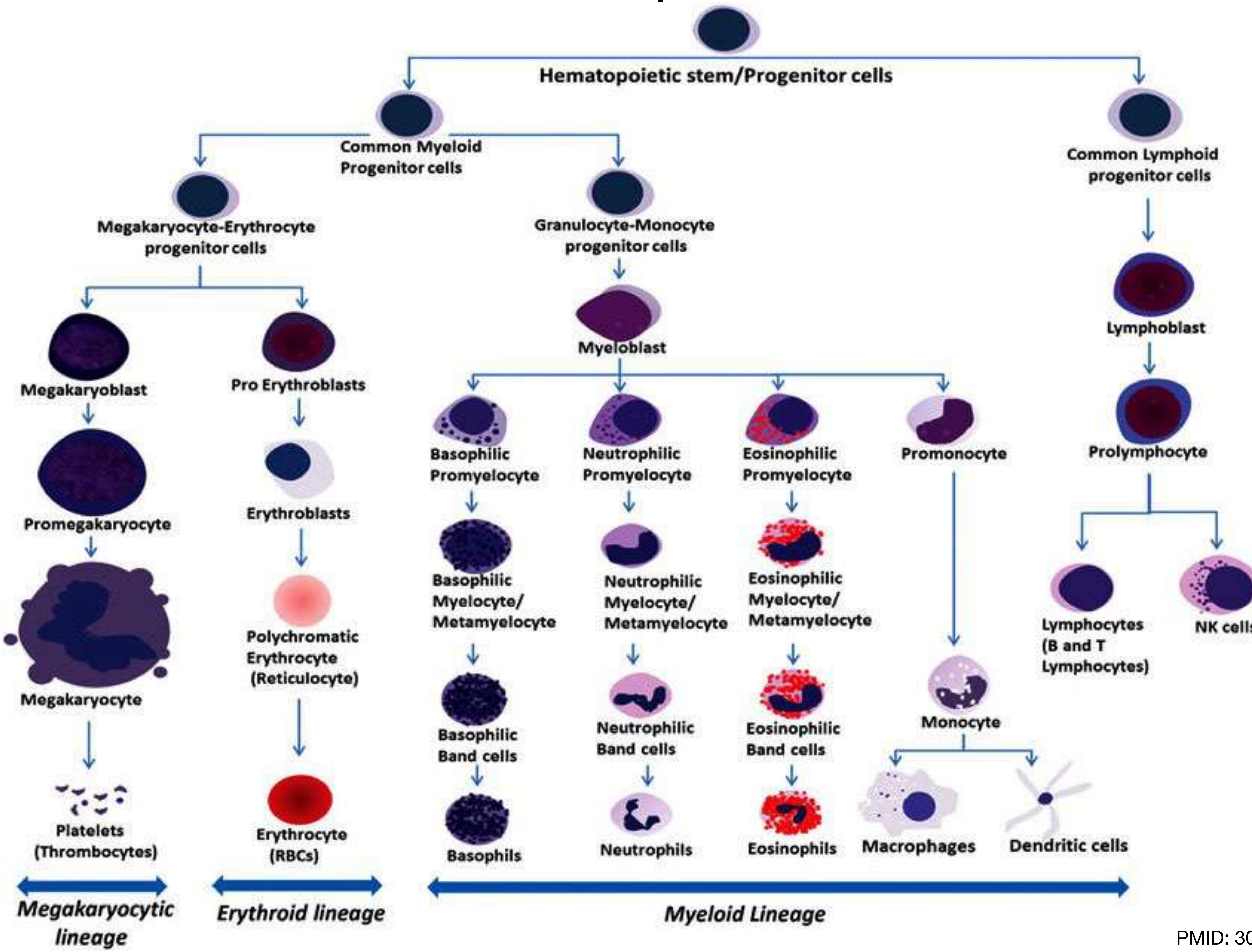


Figure 43.2 of Campbell's Biology: a global approach

Haematopoiesis



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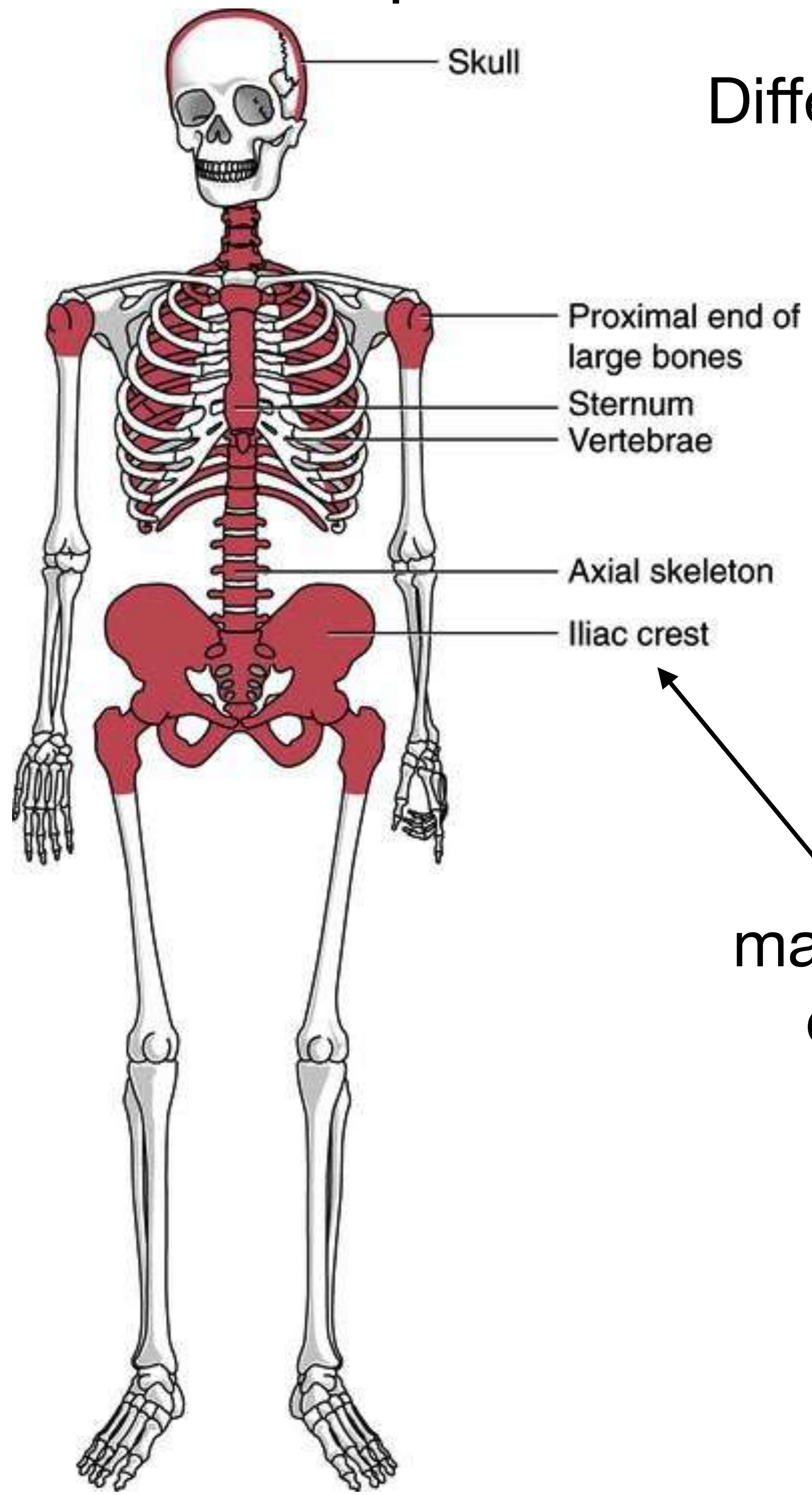
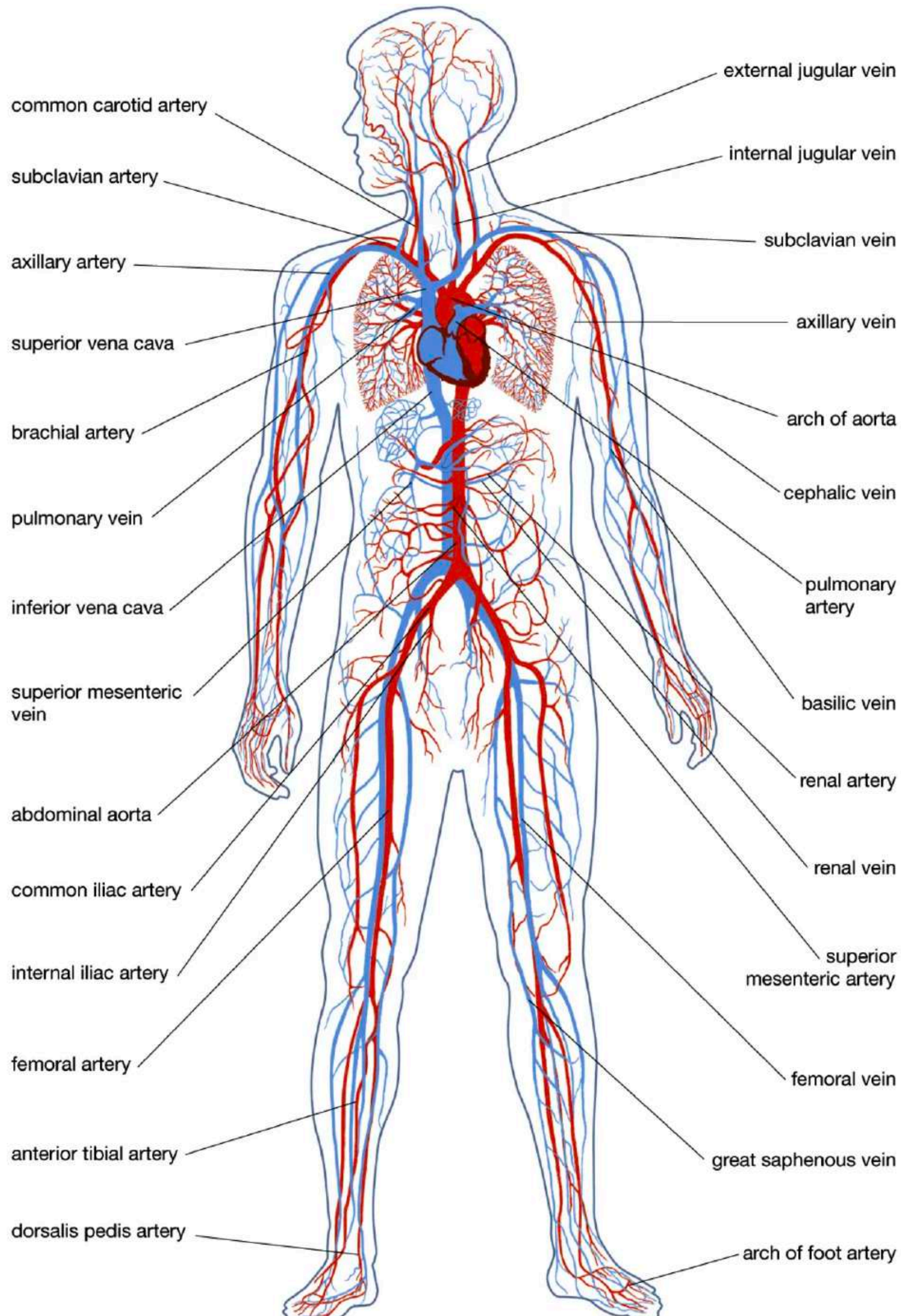
Where does haematopoiesis occur?



Different in children

Usually bone marrow biopsies are done from here

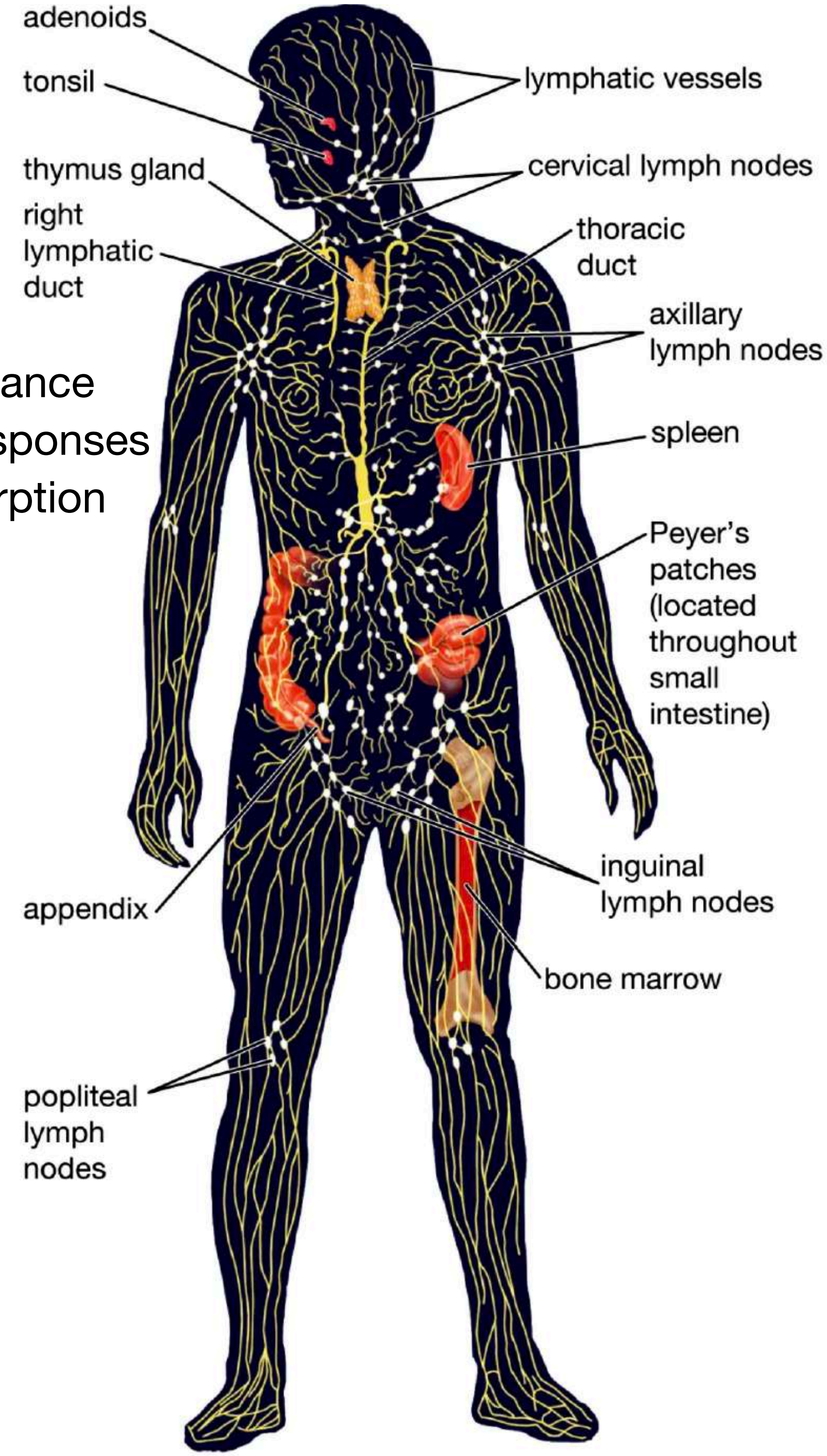
Circulatory system vs Lymphatic system



Fluid balance

Immune responses

Fat absorption



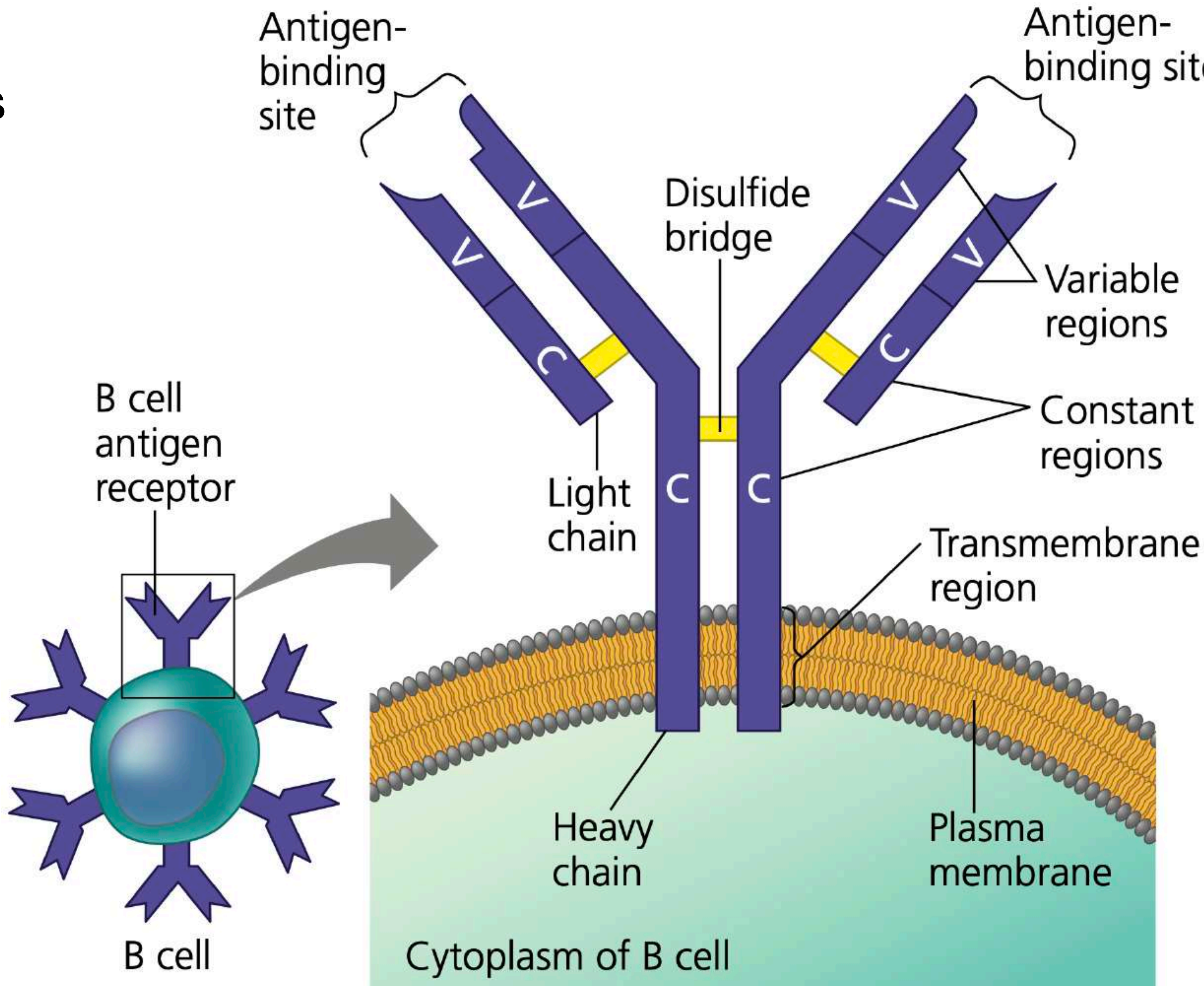
Adaptive or Acquired Immunity

- The adaptive or acquired immunity relies on T cells and B cells
- Mature B cell Mature T cell lymphocytes originate from stem cells in the bone marrow.
- Some migrate from the bone marrow to the thymus, an organ in the thoracic cavity above the heart. These lymphocytes mature into T cells.
- Lymphocytes that remain and mature in the bone marrow develop as B cells.

- Any substance that elicits a B or T cell response is called an antigen.
- In adaptive immunity, recognition occurs when a B cell or T cell binds to an antigen, such as a bacterial or viral protein
- The small, accessible portion of an antigen that binds to an antigen receptor is called an epitope
- The recognition of B or T cell with an antigen occurs via a protein called an antigen receptor.
- Although cells of the immune system produce millions of different antigen receptors, all of the antigen receptors made by a single B or T cell are identical = clonal expansion

Antigen receptor on B cells

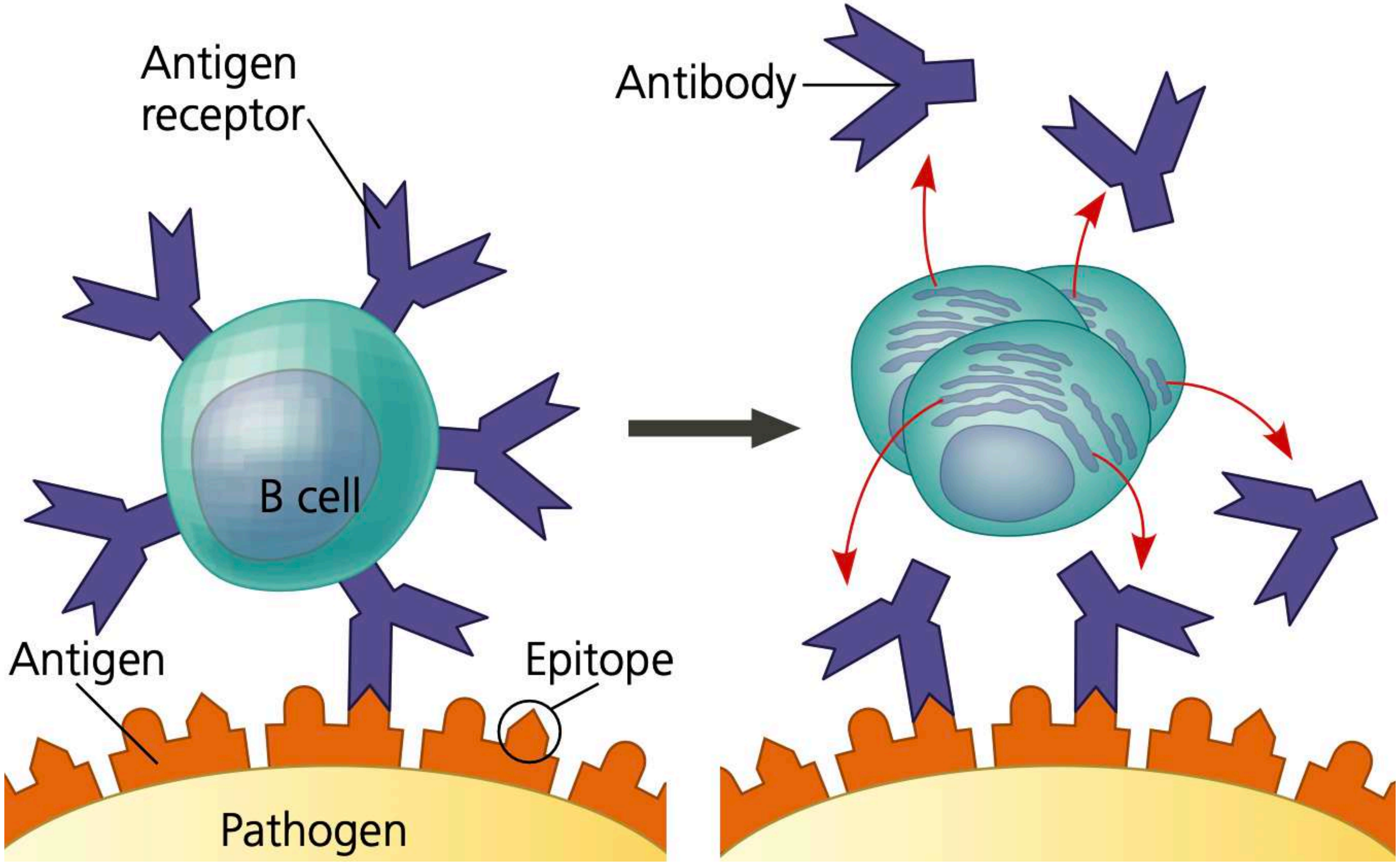
- Two identical heavy chains and two identical light chains, with disulfide bridges linking the chains together
- The light and heavy chains each have a constant (C) region, where amino acid sequences vary little among the receptors on different B cells
- Within the two tips of the Y shape each chain has a variable (V) region, so named because its amino acid sequence varies extensively from one B cell to another
- Together, parts of a heavy-chain V region and a light-chain V region form an asymmetric binding site for an antigen



Each B cell antigen receptor has two identical antigen binding sites

Figure 43.9 of Campbell's Biology: a global approach

Antigen receptor on B cells and antibodies

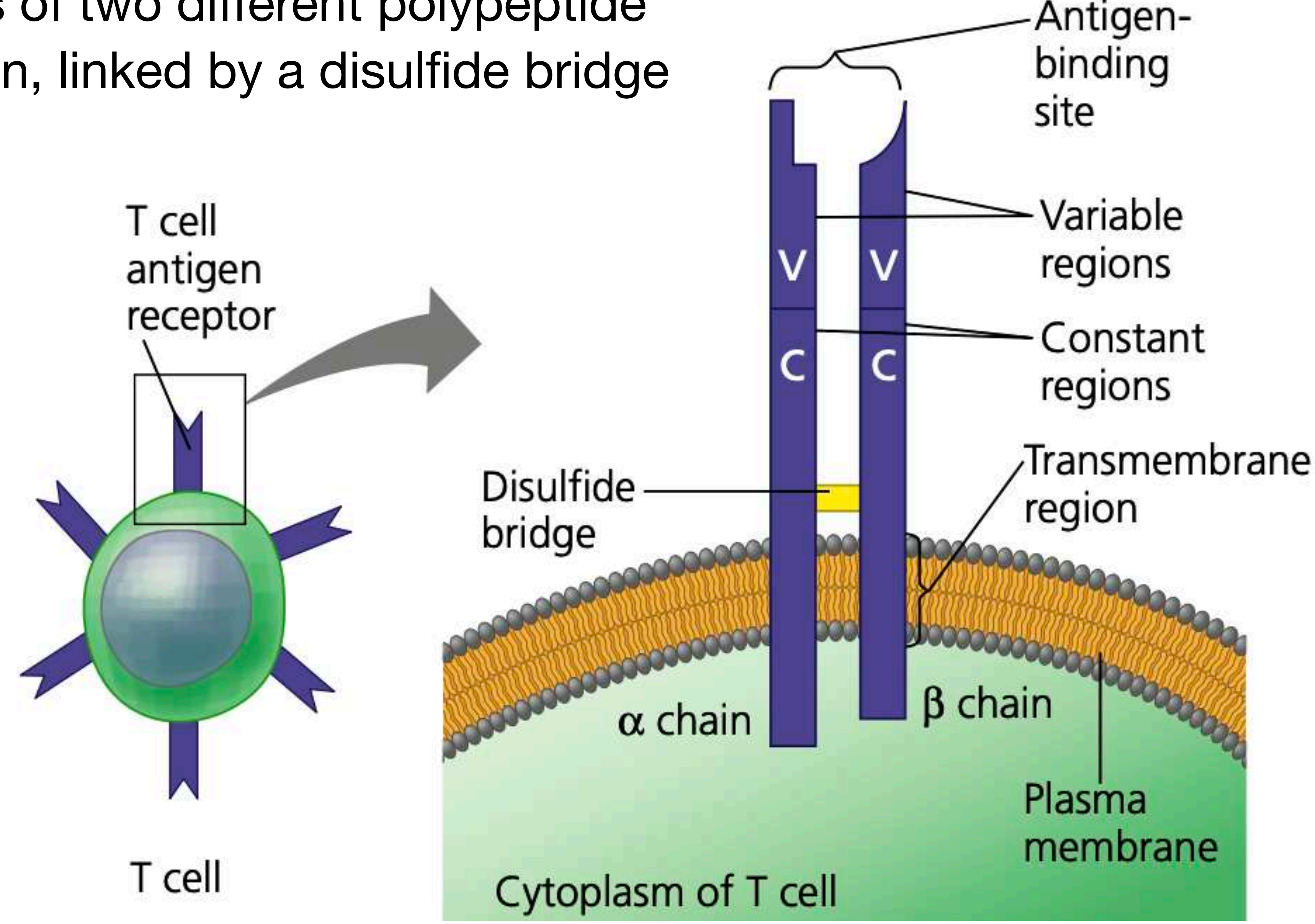


- Binding of a B cell antigen receptor to an antigen is an early step in B cell activation
- Activated B cells produce a secreted form of the antigen receptor
- This secreted protein is called an antibody
- Antibodies have the same Y-shaped structure as B cell antigen receptors but are secreted rather than membrane bound
- Antibodies can bind to antigens on the surface of pathogens or free in body fluids
- Different antibodies can be made against different antigens of the same pathogen

Figure 43.10 of Campbell's Biology: a global approach

Antigen receptor on T cells

- T cell antigen receptor consists of two different polypeptide chains, an α chain and a β chain, linked by a disulfide bridge
- At the outer tip of the molecule, the variable (V) regions of the α and β chains together form a single antigen-binding site
- The remainder of the molecule is made up of the constant (C) regions



- T cells antigen receptors bind only to fragments of antigens displayed, or presented, on the surface of host cells
- The host protein that displays the antigen fragment on the cell surface is called a major histocompatibility complex (MHC) molecule

Figure 43.11 of Campbell's Biology: a global approach

Antigen receptor on T cells

- Recognition of protein antigens by T cells begins when a pathogen or part of a pathogen either infects or is taken in by a host cell
- Inside the host cell, enzymes cleave the antigen into smaller peptides.
- Each peptide, called an antigen fragment, then binds to an MHC molecule inside the cell
- MHC molecule and bound antigen fragment move to the cell surface results in antigen presentation

Antigen presentation = advertising the fact that a host cell contains a foreign substance

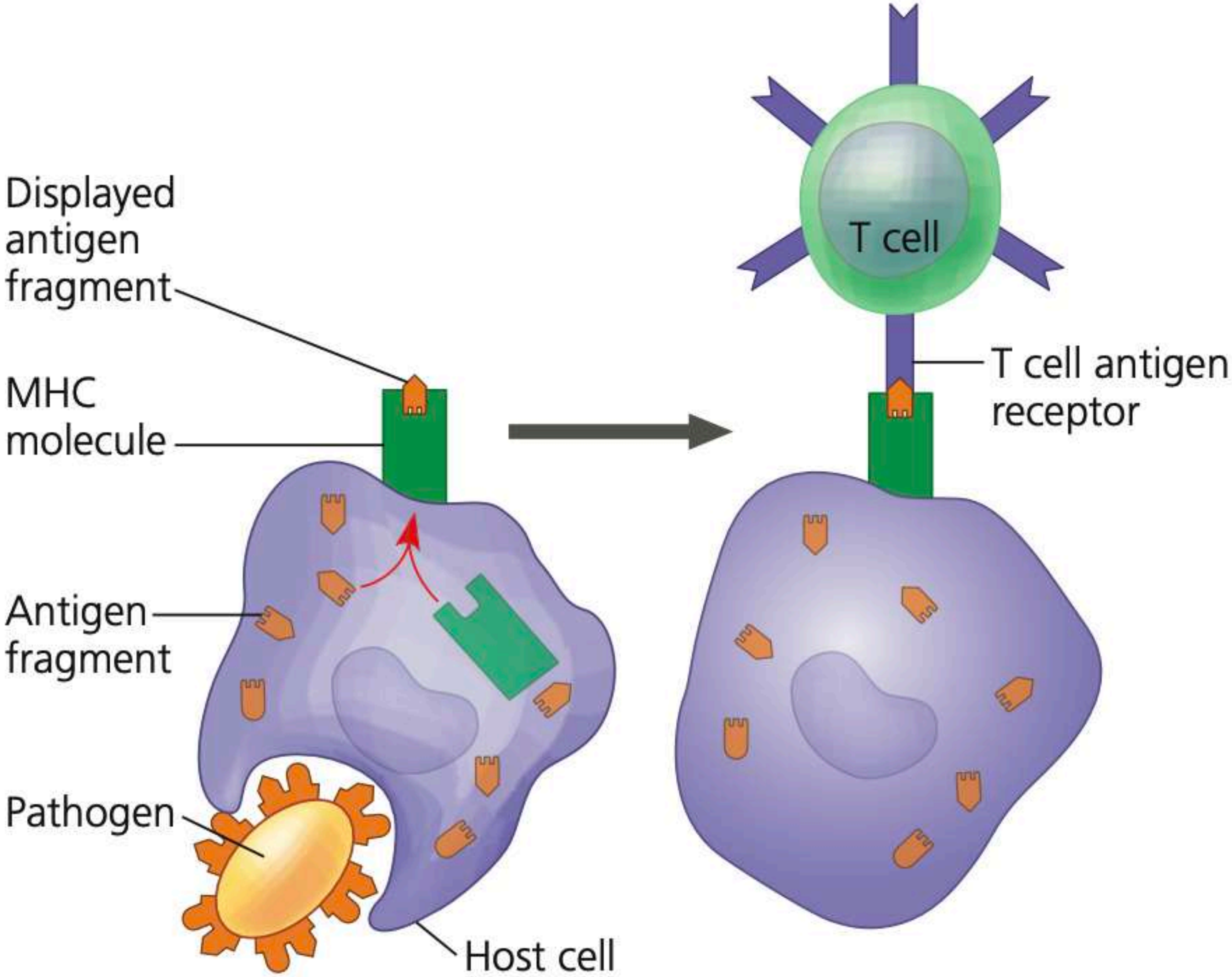


Figure 43.12 of Campbell's Biology: a global approach

Adaptive or Acquired Immunity

- Each person makes more than 1 million different B cell antigen receptors and 10 million different T cell antigen receptors
- An antigen is presented to a steady stream of lymphocytes in the lymph nodes until a match is made
- A successful match between an antigen receptor and an epitope initiates events that "activate" the lymphocyte bearing the receptor
- Once activated, a B cell or T cell undergoes multiple cell divisions.
- For each activated cell, the result of this proliferation is a clone = a population of cells that are identical to the original cell.
- Some cells from this clone become effector cells, short-lived cells that act immediately against the antigen and eliminate it.
- The effector forms of B cells are plasma cells, which secrete antibodies.
- The effector forms of T cells are helper T cells and cytotoxic T cells
- The remaining cells in the clone become memory cells, long-lived cells that can give rise to effector cells if the same antigen is encountered later in the animal's life
- This immunological memory is responsible for long-term protection against additional exposures

Adaptive or Acquired Immunity memory

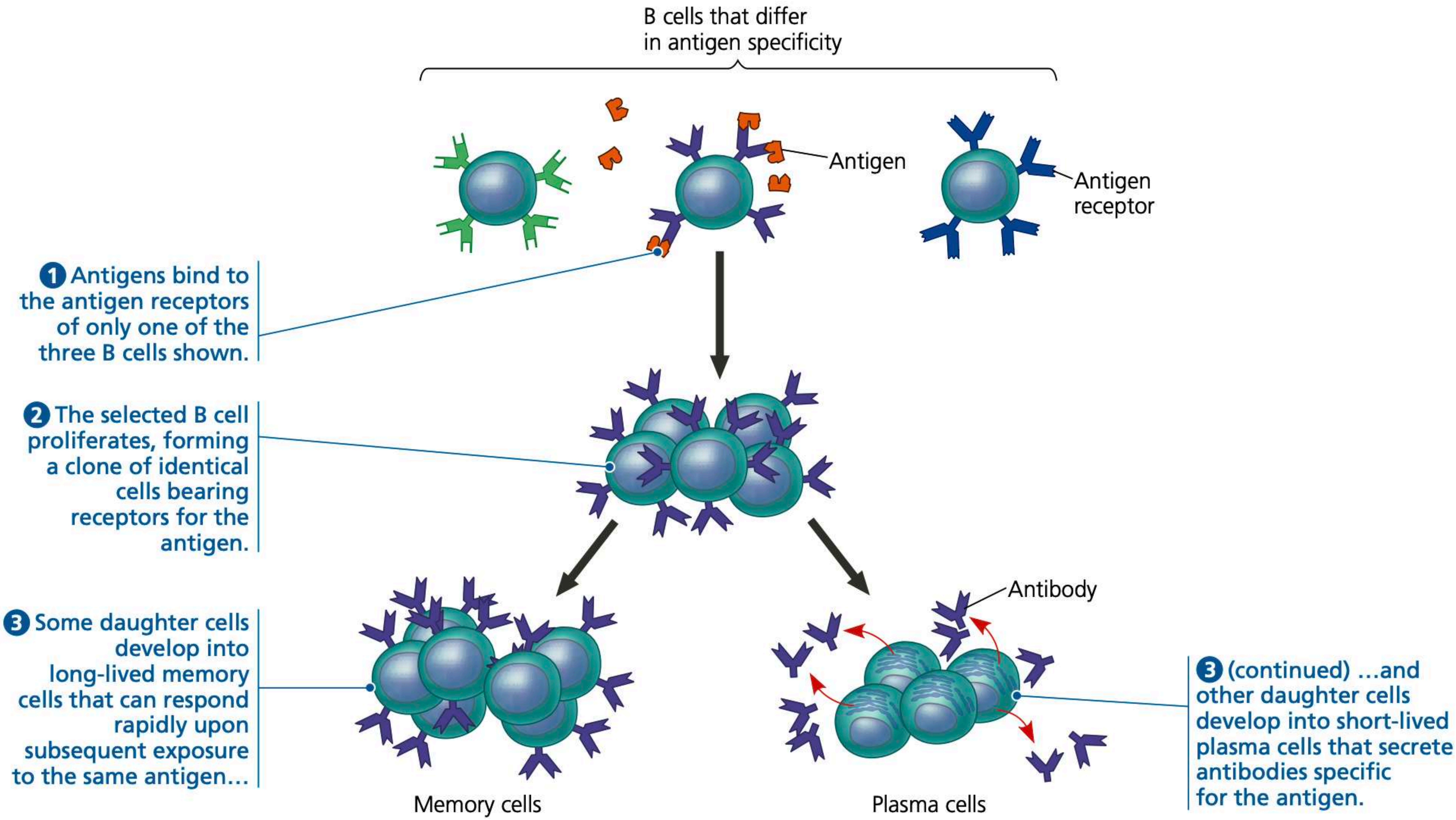
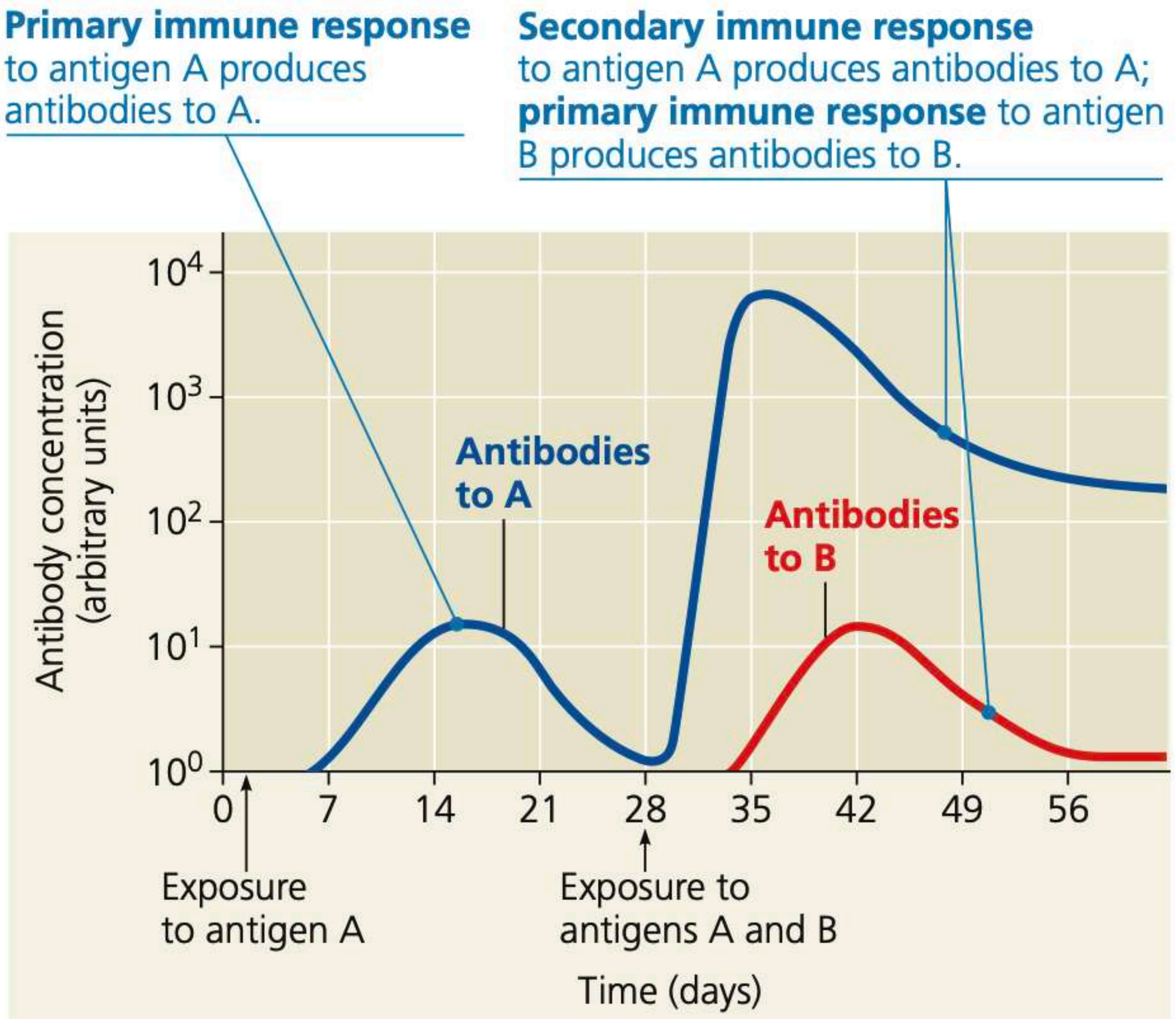


Figure 43.14 of Campbell's Biology: a global approach

Adaptive or Acquired Immunity memory



Immunological memory is specific to an antigen and does not affect responses to other antigens

Figure 43.15 of Campbell's Biology: a global approach

Antibodies do not actually kill pathogens, but by binding to antigens, they interfere with pathogen activity or mark pathogens in various ways for inactivation or destruction

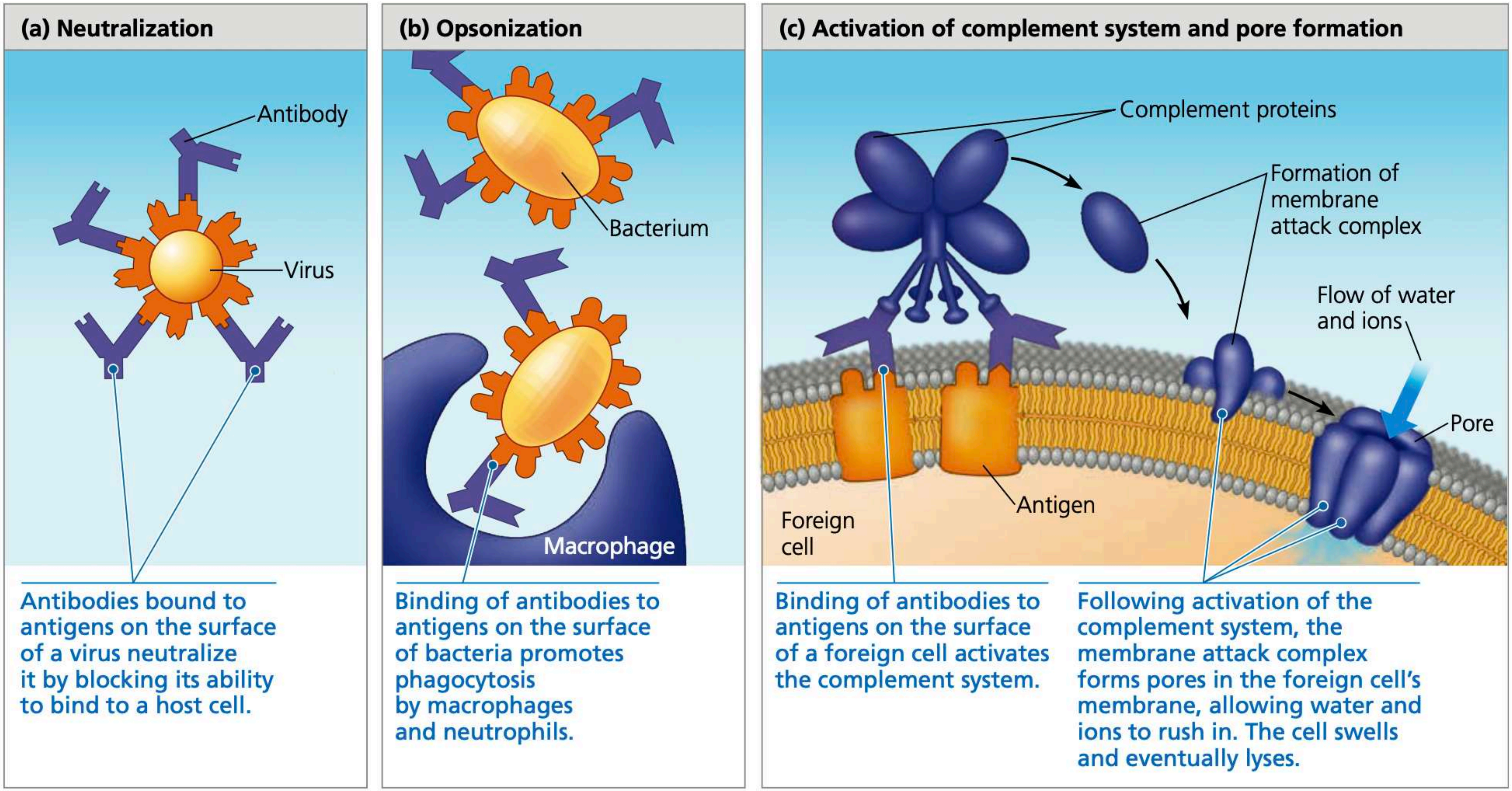


Figure 43.19 of Campbell's Biology: a global approach